

An Analysis of Image captioning

Image captioning

Image captioning -- Part 1

Chatbots Another area where NLG has been widely applied is automated dialogue systems, frequently in the form of chatbots. A chatbot or chatterbot is a software application used to conduct an on-line chat conversation via text or text-to-speech, in lieu of providing direct contact with a live human agent. While natural language processing (NLP) techniques are applied in deciphering human input, NLG informs the output part of the chatbot algorithms in facilitating real-time dialogues. Early chatbot systems, including Cleverbot created by Rollo Carpenter in 1988 and published in 1997, reply to questions by identifying how a human has responded to the same question in a conversation database using information retrieval (IR) techniques. Modern chatbot systems predominantly rely on machine learning (ML) models, such as sequence-to-sequence learning and reinforcement learning to generate natural language output. Hybrid models have also been explored. For example, the Alibaba shopping assistant first uses an IR approach to retrieve the best candidates from the knowledge base, then uses the ML-driven seq2seq model re-rank the candidate responses and generate the answer.

Image captioning -- Part 2

Creative writing and computational humor Creative language generation by NLG has been hypothesized since the field's origins. A recent pioneer in the area is Phillip Parker, who has developed an arsenal of algorithms capable of automatically generating textbooks, crossword puzzles, poems and books on topics ranging from bookbinding to cataracts. The advent of large pretrained transformer-based language models such as GPT-3 has also enabled breakthroughs, with such models demonstrating recognizable ability for creating-writing tasks. A related area of NLG application is computational humor production. JAPE (Joke Analysis and Production Engine) is one of the earliest large, automated humor production systems that uses a hand-coded template-based approach to create punning riddles for children. HAHAAcronym creates humorous reinterpretations of any given acronym, as well as proposing new fitting acronyms given some keywords. Despite progresses, many challenges remain in producing automated creative and humorous content that rival human output. In an experiment for generating satirical headlines, outputs of their best BERT-based model were perceived as funny 9.4% of the time (while real headlines from The Onion were 38.4%) and a GPT-2 model fine-tuned on satirical headlines achieved 6.9%. It has been pointed out that two main issues with humor-generation systems are the lack of annotated data sets and the lack of formal evaluation methods, which could be applicable to other creative content generation.

Some have argued relative to other applications, there has been a lack of attention to creative aspects of language production within NLG. NLG researchers stand to benefit from insights into what constitutes creative language production, as well as structural features of narrative that have the potential to improve NLG output even in data-to-text systems.

Image captioning -- Part 3

Image captioning Over the past few years, there has been an increased interest in automatically generating captions for images, as part of a broader endeavor to investigate the interface between vision and language. A case of data-to-text generation, the algorithm of image captioning (or automatic image description) involves taking an image, analyzing its visual content, and generating a textual description (typically a sentence) that verbalizes the most prominent aspects of the image. An image captioning system involves two sub-tasks. In Image Analysis, features and attributes of an image are detected and labelled, before mapping these outputs to linguistic structures. Recent research utilizes deep learning approaches through features from a pre-trained convolutional neural network such as AlexNet, VGG or Caffe, where caption generators use an activation layer from the pre-trained network as their input features. Text Generation, the second task, is performed using a wide range of techniques. For example, in the Midge system, input images are represented as triples consisting of object/stuff detections, action/pose detections and spatial relations. These are subsequently mapped to <noun, verb, preposition> triples and realized using a tree substitution grammar. A common method in image captioning is to use a vision model (such as a ResNet) to encode an image into a vector, then use a language model (such as an RNN) to decode the vector into a caption. Despite advancements, challenges and opportunities remain in image capturing research. Notwithstanding the recent introduction of Flickr30K, MS COCO and other large datasets have enabled the training of more complex models such as neural networks, it has been argued that research in image captioning could benefit from larger and diversified datasets. Designing automatic measures that can mimic human judgments in evaluating the suitability of image descriptions is another need in the area. Other open challenges include visual question-answering (VQA), as well as the construction and evaluation multilingual repositories for image description.

Image captioning -- Part 4

Pollen counts are expected to remain high at level 6 over most of Scotland, and even level 7 in the south east. The only relief is in the Northern Isles and far northeast of mainland Scotland with

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medium levels of pollen count.

Image captioning -- Part 5

Further reading Dale, Robert; Reiter, Ehud (2000). Building natural language generation systems. Cambridge, UK: Cambridge University Press. ISBN 978-0-521-02451-8. Evans, Roger; Piwek, Paul; Cahill, Lynne (2002). What is NLG?. INLG2002. New York, US. paper Gatt, Albert; Krahmer, Emiel (2018). "Survey of the State of the Art in Natural Language Generation: Core tasks, applications and evaluation". Journal of Artificial Intelligence Research. 61: 65–170. arXiv:1703.09902. doi:10.1613/jair.5477. S2CID 16946362. Reiter, Ehud (2018-01-16). "How do I Learn about NLG?".

Image captioning -- Part 6

Automatic report generation From a commercial perspective, the most successful NLG applications have been data-to-text systems which generate textual summaries of databases and data sets; these systems usually perform data analysis as well as text generation. Research has shown that textual summaries can be more effective than graphs and other visuals for decision support, and that computer-generated texts can be superior (from the reader's perspective) to human-written texts. The first commercial data-to-text systems produced weather forecasts from weather data. The earliest such system to be deployed was FoG, which was used by Environment Canada to generate weather forecasts in French and English in the early 1990s. The success of FoG triggered other work, both research and commercial. Recent applications include the UK Met Office's text-enhanced forecast. Data-to-text systems have since been applied in a range of settings. Following the minor earthquake near Beverly Hills, California on March 17, 2014, The Los Angeles Times reported details about the time, location and strength of the quake within 3 minutes of the event. This report was automatically generated by a 'robo-journalist', which converted the incoming data into text via a preset template. Currently there is considerable commercial interest in using NLG to summarise financial and business data. Indeed, Gartner has said that NLG will become a standard feature of 90% of modern BI and analytics platforms. NLG is also being used commercially in automated journalism, chatbots, generating product descriptions for e-commerce sites, summarising medical records, and enhancing accessibility (for example by describing graphs and data sets to blind people). An example of an interactive use of NLG is the WYSIWYM framework. It stands for What you see is what you meant and allows users to see and manipulate the continuously rendered view (NLG output) of an underlying formal language document (NLG input), thereby editing the

formal language without learning it. Looking ahead, the current progress in data-to-text generation paves the way for tailoring texts to specific audiences. For example, data from babies in neonatal care can be converted into text differently in a clinical setting, with different levels of technical detail and explanatory language, depending on intended recipient of the text (doctor, nurse, patient). The same idea can be applied in a sports setting, with different reports generated for fans of specific teams.